Analytical Chemistry Cumulative Exam (February 10, 2023)

Announced Topic: Calibration Curves and Least Squares Analysis

1. (12 points) Name (a) three different kinds of calibration curves, and (b) the circumstances under which each should be used.

2. (9 points) Name one assumption that is made when using the method of least squares for two-dimensional data when generating the best-fit line to data points?

3. (10 points) What is the *coefficient of determination* (also known as the *square of the correlation coefficient*)? What does it tell you about a calibration curve?

4. (18 points) An analysis of an unknown amount of copper in a river sample using atomic absorption spectroscopy at a certain wavelength was performed. **10.00 mL** aliquots of river sample were added to each of a set of flasks. Then, increasing volumes of a solution of copper standard were added to the flasks (one of the flasks contained no added copper standard). The solutions in the flasks were then diluted to the same volume of **100.00 mL**. The absorbance value, A, for each diluted solution was plotted as a function of final copper concentration of standard added, x.
The equation of the least squares best fit line through calibration data was obtained to be:

$$\mathbf{A = 0.521\ (mL/\mu g)\ x + 0.0500}$$

What is the concentration of the unknown in the *original* sample?

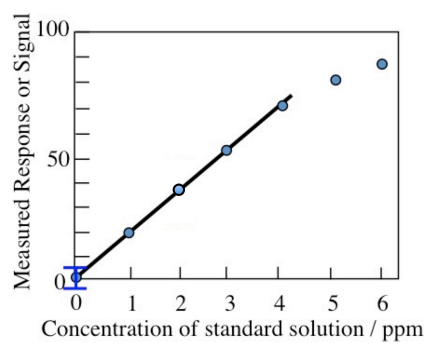
5. (18 points) A researcher was attempting to quantify the amount of a pesticide in kale using gas chromatography mass spectrometry (GC/MS). The researcher eventually used the ***internal standard calibration*** method with chloroform as the internal standard.

The response factor of the mass spectrometer (based on peak areas and concentrations in terms of mg/L) for the pesticide to chloroform is **0.218**.

Chloroform was added to a kale extract so that the final chloroform concentration was **0.513 mg/L**. This solution was analyzed by GC/MS, producing peak areas of **8021 for the unknown pesticide** and **14008 for the chloroform**. Calculate the pesticide concentration in the kale extract.

6. (18 points) An ***external calibration*** curve with **5 points** was produced to determine the concentration of dopamine in a particular region of rat brain with electrochemical detection at a carbon fiber electrode. The least squares best fit line to the points in the calibration plot was
y = 1.052 x + 0.058, where y is current in nA and x is concentration in nanomolar (nM). The value of the regression about the line was **0.395**. A current of **8.0 nA** was obtained for an unknown using **only one analysis**. What is the concentration of the analyte in the unknown ± standard deviation of that concentration? *(Note that* $\bar{y} = 67$*, and* $S_{xx} = \sum (x_i - \bar{x})^2 = 7114.8$*).*

7. (15 points) Determine the value of each of the following parameters from the plot at the right. **Calculate your best estimate and provide units.** (Note: use the unit "signal" for the measured response and "ppm" for the concentration).
Show work where possible.
What is the linear dynamic range?_____
What is the calibration sensitivity? _____
What is the detection limit in terms of concentration units?_____



Equations that you might find useful on this cumulative exam.

Some Definitions:

x_i = x-coordinate of a data point (x_i,y_i) in the calibration curve.

y_i = y-coordinate of a data point (x_i,y_i) in the calibration curve.

$\bar{x}$ = average of the x-coordinates of the data points

$\bar{y}$ = average of the y-coordinates of the data points

N = n = number of points in the calibration curve

m = slope

b = y-intercept

s_m = standard deviation of the slope

s_b = standard deviation of the y-intercept

$s_y = s_r$ = Standard deviation about the regression (also known as the standard deviation of deviations of the y-coordinates from the best fit line)

$s_x = s_c$ = standard deviation for concentration of an unknown obtained from the calibration curve

k = M = number of replicates performed on the unknown

$\overline{y_c} = y$ = average measured signal for the unknown from the k or M replicates

Notation used by Skoog.

$$S_{xx} = \Sigma\left(x_i - \bar{x}\right)^2 = \Sigma x_i^2 - \frac{\left(\Sigma x_i\right)^2}{N}$$

$$S_{yy} = \Sigma\left(y_i - \bar{y}\right)^2 = \Sigma y_i^2 - \frac{\left(\Sigma y_i\right)^2}{N}$$

$$S_{xy} = \Sigma\left(x_i - \bar{x}\right)\left(y_i - \bar{y}\right) = \Sigma x_i y_i - \frac{\Sigma x_i \Sigma y_i}{N}$$

$$m = \frac{S_{xy}}{S_{xx}}$$

$$b = \bar{y} - m\,\bar{x}$$

$$s_r = \sqrt{\frac{S_{yy} - m^2 S_{xx}}{N - 2}}$$

$$s_m = \sqrt{\frac{s_r^2}{S_{xx}}}$$

$$s_b = s_r \sqrt{\frac{\Sigma x_i^2}{N \Sigma x_i^2 - \left(\Sigma x_i\right)^2}} = s_r \sqrt{\frac{1}{N - \left(\Sigma x_i\right)^2 / \Sigma x_i^2}}$$

$$s_c = \frac{s_r}{m} \sqrt{\frac{1}{M} + \frac{1}{N} + \frac{\left(\overline{y_c} - \bar{y}\right)^2}{m^2 S_{xx}}}$$

$$R^2 = 1 - (SS_{resid} / SS_{tot})$$

$$SS_{resid} = \Sigma\ [y_i - (b + m\ x_i)]^2$$

$$SS_{tot} = S_{yy}$$

$$SS_{regr} = SS_{tot} - SS_{resid}$$

$$R^2 = (SS_{regr} / SS_{tot})$$

Notation used by Harris.

$$D = \begin{vmatrix} \sum\left(x_i^2\right) & \sum x_i \\ \sum x_i & n \end{vmatrix} = \left(\sum\left(x_i^2\right)\times n\right) - \left(\sum x_i \times \sum x_i\right)$$

$$S_{r_i^2} = S_{d_i^2} = \sum\left(d_i^2\right) = \sum\left(y_i - (mx_i + b)\right)^2$$

$$m = \begin{vmatrix} \sum\left(x_i y_i\right) & \sum x_i \\ \sum y_i & n \end{vmatrix} \div D$$

$$b = \begin{vmatrix} \sum\left(x_i^2\right) & \sum\left(x_i y_i\right) \\ \sum x_i & \sum y_i \end{vmatrix} \div D$$

$$s_y = \sqrt{\frac{\sum\left(d_i^2\right)}{n-2}}$$

$$s_m = \sqrt{\frac{s_y^2 \mathrm{n}}{\mathrm{D}}}$$

$$s_b = \sqrt{\frac{s_y^2 \sum\left(x_i^2\right)}{D}}$$

uncertainty in x (= s_x) = $\frac{s_y}{|m|}\sqrt{\frac{1}{k} + \frac{1}{n} + \frac{\left(y - \bar{y}\right)^2}{m^2 \sum\left(x_i - \bar{x}\right)^2}}$

$$R^2 = \frac{\left[\sum\left(x_i - \bar{x}\right)\left(y_i - \bar{y}\right)\right]^2}{\sum\left(x_i - \bar{x}\right)^2 \sum\left(y_i - \bar{y}\right)^2}$$